

UWC Anatomical Landmark Detection User Manual

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Problem Statement	3
Ethical Considerations and Decisions.....	3
Privacy and Data Security	3
Ethical AI Use	3
Data Retention.....	4
Data	4
Datasets	4
Data Quality	4
Data Processing.....	5
Modelling	7
Model Architectures	7
Model Iterations on the South African data	8
Model Selection	11
Limitations	12
Package	14
Requirements	14
Installation	15
Cephalopy Package	15
API Endpoints	16
Application	20
Set-up	19
Functionalities	20
Future Recommendations	24
Data Recommendations	24
Deployment Recommendations.....	25
Application Recommendations	25
References	26

Problem Statement

Cephalometric analysis requires plotting several landmarks for linear and angular measurements to assess the growth and development of a skull (Ghondasra et al, 2023). Manual cephalometric analysis can introduce subjectivity and inconsistency in the landmark points and costs time (Ye et al, 2023). The introduction of AI in this task may remove inconsistency, reduce random errors, and save time. Technology already exists for this task, mainly based on European and Asian datasets (Indermun et al., 2023). Due to the extensive variety in African diversity, the University of Western Cape (UWC) faces an issue where these existing tools do not provide optimal results in dental environments in South Africa. To ensure greater, faster, and more efficient outcomes for South Africa's broader diversity in population, the client has requested the training of an anatomical landmark detection tool specifically trained on South African datasets. This tool should target industry-standard detection accuracy of under two millimetres (Wang et al., 2016).

Ethical Considerations and Decisions

In medical systems, especially AI-based systems, research into the ethicality of developed work is a very important step to take. Here it is important to understand, analyse and address any ethical concerns that might occur with the usage of the data and outcome of systems. This section will describe and specify the ethical research and decisions related to data required for this project. A full report is written on a separate Ethical Considerations and Decisions document¹

Privacy and Data Security

This system processes anonymised cephalometric X-ray images only. All patient-identifiable information must be removed from metadata before upload. Uploaded images are stored securely using encryption through cloud storage with restricted access. The system is designed to comply with GDPR requirements and medical data protection standards.

Ethical AI Use

This system is designed to support, but not replace, clinical judgement. Results should be reviewed by qualified professionals. The system was trained on diverse datasets and

¹ Ethical Considerations and Decisions: https://edubuas.sharepoint.com/:w:/t/LandmarkDetection/IQD-eU_ikviiR5SVL3q1Mz95AfU33DAIABOZaHD2SEvpHgs?e=H1ROiv

includes South African data to improve accuracy for local populations. Performance may vary, and users should exercise professional discretion when interpreting results.

Data Retention

Uploaded images and predictions are retained for quality improvements and retraining for developed solution. Data is stored securely for a minimum of 5 years in compliance with research standards.

Data

Datasets

To avoid bias in the model training process, datasets from multiple different sources have been selected and used for model training. The external datasets we have selected: Aariz (Khalid et al., 2022), PKU (Zeng et al., 2020), and a cleaned dataset taken from a 2019 Kaggle dataset, of which the data originates from the ISBI 2015 Grand Challenge dataset (Wang et al., 2015).

Before the inclusion of the client's South African data, this initial dataset included 1502 cephalometric radiographs with corresponding annotations. The Aariz dataset contains 67% of that total data, with 1000 images and annotations; geographical origin is not confirmed. The PKU dataset is the smallest, containing only 102 cephalograms originating from Peking, China. The ISBI 2015 Grand Challenge dataset contains 400 images from Taiwanese origin. Finally, we received 212 images from UWC, which are of South African origin.

Data Quality

Quality assessment has been conducted on the datasets through python, aiming to verify the quality of completeness, consistency, validity, and uniqueness of the datasets. As seen in Table 1, the combined dataset achieved 100% completeness across all three subsets, where 1502 valid image and annotation pairs were found, with no missing or incomplete records.

Dataset	Total Images	Total Annotations	Matched Pairs	Missing Data	Incomplete Landmarks	Valid Samples	Completeness
Aariz	1,000	1,000	1,000	0	0	1,000	100%
PKU	102	102	102	0	0	102	100%
ISBI	400	400	400	0	0	400	100%
Total	1,502	1,502	1,502	0	0	1,502	100%

Table 1: Data Completeness Assessment Across Datasets

Data consistency of landmark point coordinates has been assessed through implementation of 3 analyses:

1. Check if coordinates fall within image pixel boundaries.
2. Analysis of intervals across the dataset of each landmark.
3. Calculating distances between critical landmark pairs to ensure clinically plausible measurements.

Along with the above analyses, five critical anatomical measurements were also evaluated. These evaluations serve as anatomical validation gates, where unrealistic distances would indicate annotation errors;

1. N-Me (Nasion to Menton; total facial height)
2. A-B (A-point to B-point; jaw relationship)
3. S-N (Sella to Nasion; cranial base length)
4. Go-Me (Gonion to Menton; mandibular body length)
5. UIT-LIT (Upper to Lower incisor tip; incisor relationship)

The outcome of the above analyses indicated 100% data consistency, where no extreme outliers or anatomically impossible distances could be detected. Validity also scored 100%, as all annotations fell within corresponding image dimensions.

The uniqueness of the dataset was measured through implementation of an MD5 hashing algorithm, which would create unique fingerprints for each image file. With this algorithm, 11 duplicate images were identified, indicating 98.4% uniqueness.

Finally, the Data Quality KPI was calculated, where the outcome was 99.8%. This indicates that most of the dataset is of high quality, only containing 11 duplicate images, which were afterwards addressed correctly by removing the duplicates.

Data Processing

Before starting the modelling process, it is important to process the data to make everything consistent and ready for model training. Through the data processing phase, many important decisions were made, and experiments were conducted, so all successful and currently deployed methods and steps will be explained here.

Due to all datasets containing inconsistent file names and formats, preprocessing started with ensuring file consistency. All annotations were converted to JavaScript Object Notation (JSON) files, containing the landmarks in a consistent order. All images were converted to PNG files, and the image and annotations were renamed to follow a

consistent naming convention: '{datasetname}_{imageID}_cephalogram.png' for images and '{datasetname}_{annotationID}_annotation.json' for the annotations.

After this step, cephalograms are contrasted through histogram distribution matching to a reference image: PKU's '54.bmp'. This change ensures proper and consistent pixel values across all images, helping with the next processing task: **Temporomandibular Joint (TMJ)** detection. The TMJ is where your lower jaw connects to your skull and is the start of the region of interest in this cephalometric analysis task. Before detecting the TMJ, the left quarter of the image is examined to detect irrelevant areas on one side of the images. If more than 70% of this section is black, the region of interest would start further on in the image, so cropping will be applied.

Selecting the TMJ is done through binary thresholding with the value of 200 to make a separation of the brighter anatomical structures, followed by selecting objects closest to image centre and filtering out noise, which ensures selection of the TMJ (Figure 1).

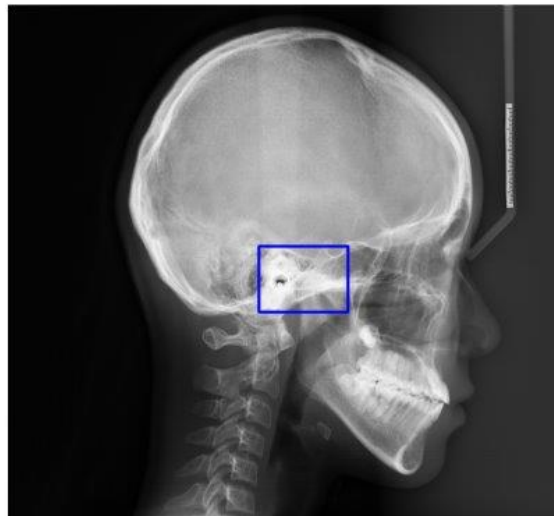


Figure 1: Temporomandibular Joint (TMJ) selection outcome.
PKU (Zeng et al., 2020)

Cropping is then based on the location of the TMJ, where the leftmost point of this object is found, and a 200-pixel margin towards the left is added as safety measure. Due to this process being prone to error, a safety check is also performed; if more than 40% of the image width is removed in this process, the cropping will be limited to only 40% to avoid over-cropping. In the end, horizontal cropping is performed on the final selected location, and 200 pixels are also always removed from the top of the image. This preserves the entire anatomical region of interest with minimal failure (Figure 2).

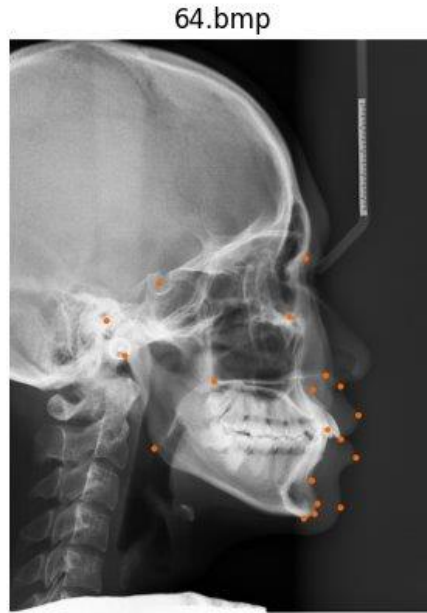


Figure 2: Image cropping outcome example
PKU (Zeng et al., 2020)

Right before training starts, the images are processed for further use inside the model. Heatmaps are generated from landmark annotations, ensuring the model receives an understanding of landmark context and increasing model explainability by bringing clear understanding of its decisions. Images are then resized to 512x512 pixels, and heatmaps are resized to 128x128 pixels, decreasing training and inference time.

Modelling

The modelling phase involved structured stages consisting of multiple architectures and training strategies to identify the optimal approach for cephalometric landmark detection on South African patient populations. This section highlights relevant model iterations, their evaluation and performances, and the decision-making process that led to the final deployed solutions.

Model Architectures

Initial Baseline: VGG-16

The first model developed was a **VGG-16 convolutional neural network** (Simonyan & Zisserman, 2015), selected as a baseline architecture due to its proven effectiveness in medical image analysis tasks. VGG-16 consists of 13 convolutional layers organized into five sequential blocks with max-pooling operations, providing hierarchical feature extraction from input radiographs. This baseline model was trained on the combined

external datasets (Aariz, PKU, ISBI 2015) at 512×512 pixel resolution, achieving a Mean Radial Error (MRE) of 2.34mm.

DeepFuse Architecture

After establishing a baseline, a **DeepFuse architecture** was implemented. While the full DeepFuse framework supports multimodal fusion across 3D imaging modalities, this implementation focuses exclusively on the 2D lateral cephalogram pathway, as CBCT volumes and digital dental models were not available in the training dataset and out of the scope. The cephalogram encoder utilizes the same VGG-16 backbone architecture as the baseline model but incorporates a **multi-scale feature fusion module** that combines features from multiple hierarchical levels within the network. This multi-scale approach enables the model to capture both fine-grained local anatomical details (edges, texture patterns around specific structures) and broader structural context (overall skull morphology, relative positioning) necessary for accurate landmark localization (Gao & Tang, 2025).

Model Iterations on the South African data

Initial Baseline evaluation on the SA data

The baseline DeepFuse model trained on external datasets (n=1,502) was evaluated on South African test data to assess population-specific performance:

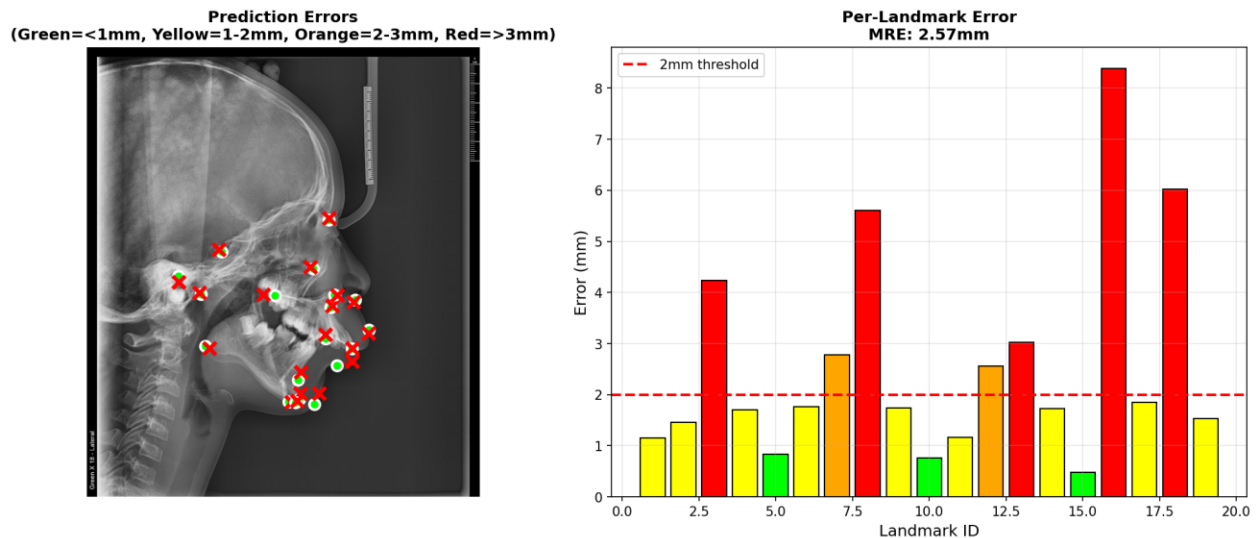


Figure 3: Baseline Performance on South African data.

Baseline Performance on SA Test Set (n=22 images):

- MRE: **2.04mm**
- SDR@2mm: **62.0%**
- SDR@3mm: 80.9%
- SDR@4mm: 87.6%

Finding: The 62.0% SDR@2mm demonstrated measurable population bias, falling substantially below the 85% clinical deployment threshold. This performance gap was anticipated based on existing commercial systems evaluated on South African populations: DentalIQ achieved only 66.5% accuracy and WebCeph 57.2% accuracy, both falling substantially short of clinical requirements (Indermun et al., 2023).

Fine-Tuning Approach (Failed Experiment)

Methodology: Initial mitigation attempts employed transfer learning, fine-tuning the pre-trained baseline model with South African training data (n=170 images) while maintaining the 512×512 resolution.

Performance Results:

- MRE: **2.41mm** (degraded from 2.04mm)
- SDR@2mm: **37.8%** (degraded from 62.0%)

Root Cause Analysis: Diagnostic analysis identified **compression ratio mismatch** as the failure mechanism. South African radiographs (2,308-2,960×2,652 pixels) experienced 50-78% higher compression when resized to 512×512 pixels compared to the original training data (1,968×2,025 pixels). This compression disparity created incompatible feature representations that prevented successful knowledge transfer, with the pre-trained weights forcing inappropriate feature expectations at the wrong spatial scale.

Conclusion: Fine-tuning degraded performance by **39% relative to baseline**, demonstrating that transfer learning fails when fundamental domain characteristics (image resolution, compression ratios) differ substantially between source and target populations.

Training from Scratch: 512×512 Resolution (SA-Only Model)

Methodology: Training exclusively from scratch on South African data at 512×512 resolution, avoiding pre-trained weight initialization.

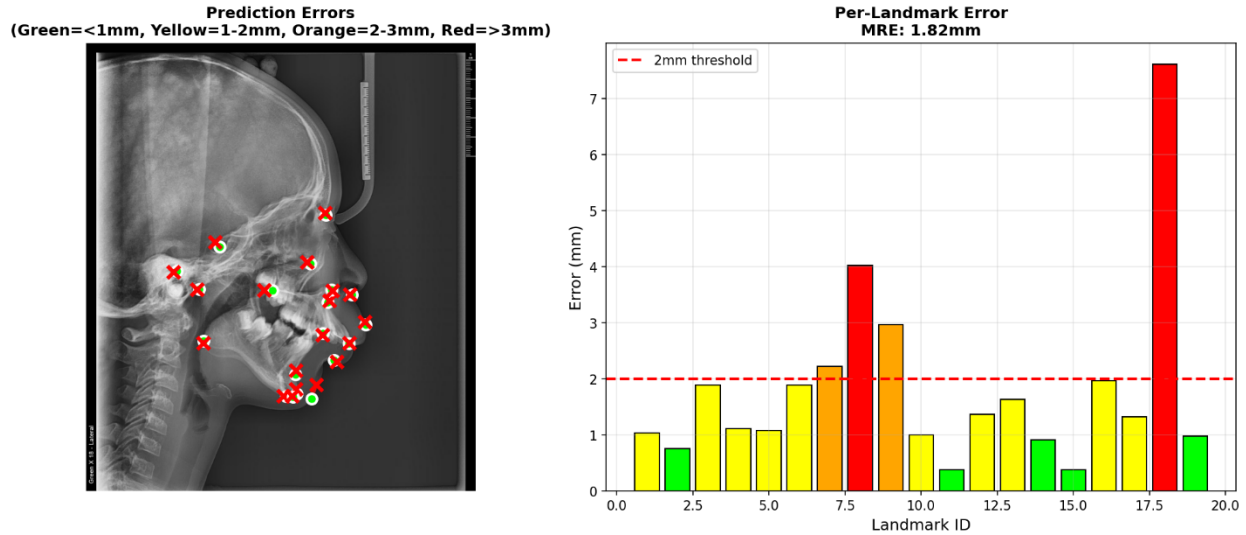


Figure 4: Performance on the same data sample.

Performance Results:

- MRE: **1.46mm**
- SDR@2mm: **81.58%**
- SDR@3mm: 91.63%
- SDR@4mm: 95.45%
- Catastrophic failures (>4mm): 4.55%

Clinical Significance: Performance approached within 3.42 percentage points of the 85% clinical deployment threshold, demonstrating substantial mitigation of population bias through population-specific training.

Training from Scratch: Combined Dataset Model (All Populations)

Methodology: Training from scratch on the complete combined dataset, including all four sources (UWC South African, Aariz, PKU, ISBI 2015) at 512×512 resolution.



Figure 5: Combined Population Model Prediction vs. Ground Truth.

Performance Results:

- MRE: **1.20mm**
- SDR@2mm: **85.8%**
- SDR@3mm: 93.6%
- SDR@4mm: 96.4%

Key Achievements:

- **41.2% relative improvement** in MRE over baseline (2.04mm → 1.20mm)
- **38.4% relative improvement** in SDR@2mm over baseline (62.0% → 85.8%)
- **Exceeded clinical deployment threshold** (85% SDR@2mm)

Clinical Significance: This model represents the first configuration to exceed the 85% clinical deployment threshold, demonstrating that training from scratch on diverse populations, including South African data, enables both population-specific accuracy and broad generalization.

Model Selection

Production Model: Combined population model trained from scratch at 512×512 resolution

Selection Rationale:

1. Exceeds clinical deployment threshold (85.8% vs. 85% required)
2. Lowest MRE achieved across all experiments (1.20mm)

3. Demonstrates generalization across diverse populations
4. Clinically deployable performance with human oversight

Limitations

Clinical Deployment Threshold Gap

Despite achieving substantial improvements over baseline architectures and commercial systems tested on South African populations, the SA-only model's 81.58% SDR@2mm remains **3.42 percentage points below** the 85% threshold recommended by Lindner et al. (2016) for fully autonomous clinical operation. The combined dataset model successfully exceeded this threshold at 85.8% SDR@2mm, demonstrating that the SA-only model's performance gap is attributable to **limited training data volume** rather than fundamental architectural limitations.

Dataset Size Constraints

Limited South African Training Data: The SA-only model was trained on only **212 South African radiographs** (170 training, 21 validation, 21 test), substantially smaller than the 1,502-image external dataset or the 1,714-image combined dataset. This constraint directly impacts:

- **Performance ceiling:** The 3.42 percentage point gap to the clinical threshold likely reflects insufficient training examples rather than model capacity limitations
- **Generalization to edge cases:** Uncommon anatomical variations, craniofacial anomalies, or pathological cases with limited training representation may produce less reliable predictions
- **Population diversity coverage:** Potential underrepresentation of the full range of South African facial morphology diversity across age groups, ethnic backgrounds, and anatomical variations

Expected improvement trajectory: The combined dataset model's superior performance (85.8% vs. 81.58% SDR@2mm) when incorporating SA data alongside larger external datasets suggests that expanding the South African training set to 400-500 images would likely enable the SA-only model to exceed the 85% clinical threshold while maintaining population-specific optimization.

Inter-Dataset Heterogeneity

Radiographic Acquisition Variability: Subtle differences in imaging protocols across datasets may introduce measurement inconsistencies:

- **Pixel spacing variations:** Different radiographic equipment and acquisition settings result in varying pixel-to-millimeter ratios across datasets (estimated range: 0.08-0.12 mm/pixel based on image dimensions and standard cephalometric proportions)
- **Resolution inconsistency:** Source images range from 1,968×2,025 pixels (external datasets) to 2,308-2,960 pixels (South African data), introducing variable information density
- **Compression effects:** Resizing to standardized 512×512 input dimensions apply different compression ratios (3-4× for external data vs. 5-6× for SA data), potentially affecting feature representation consistency

While preprocessing steps (contrast normalization, TMJ-based cropping) standardize image characteristics, residual pixel spacing differences of ±20-30% across datasets may contribute to minor systematic measurement variations. All reported MRE and SDR metrics assume uniform 0.1 mm/pixel scaling; actual pixel spacing variations could shift absolute error measurements by ±0.2-0.3mm, though relative performance comparisons between models remain valid as all models were evaluated using consistent scaling assumptions.

Impact assessment: The combined dataset model's ability to exceed clinical thresholds despite these heterogeneities suggests they introduce minor noise rather than fundamental limitations. However, future work should validate pixel spacing through radiograph calibration markers to ensure measurement accuracy across datasets.

Resolution Trade-offs

The selected **512×512 pixel resolution** optimizes for prediction stability and computational efficiency but involves trade-offs:

Detail Preservation: Higher-resolution variants (1024×1024 pixels) theoretically preserve finer anatomical details visible in original radiographs. However, empirical evaluation revealed that increased resolution paradoxically produced less stable predictions in preliminary experiments, likely due to overfitting to high-frequency noise patterns with limited training data.

Coordinate Scaling: All predictions are generated at 512×512 resolution and scaled back to original image dimensions. This introduces minor interpolation artifacts, though error analysis indicates scaling contributes negligibly to overall MRE (<0.05mm impact).

Optimal resolution selection: The 512×512 resolution represents the empirically determined balance between anatomical detail preservation and robust generalization given current dataset sizes. Larger training datasets (>1,000 SA images) may enable effective utilization of higher resolutions without overfitting.

Package

Requirements

When using the provided package and API, it is expected that the proper dependencies are installed in the used python environment. Python versions 3.10.11 up to 3.15 are compatible with the package code, with the recommendation to stick close to 3.10.11. All package dependencies are shown in Table 2 below.

Dependencies	Version
Python	>=3.10.11, <3.15
numpy	>=1.26.0
scikit-learn	>=1.5.0
tqdm	>=4.66.0
matplotlib	>=3.9.0
pillow	>=10.4.0
opencv-python	^4.10.0.88
scikit-image	^0.24.0
torch	>=2.0.0
torchvision	>=0.15.0
python-multipart	^0.0.20
reportlab	^4.2.4
python-dotenv	^1.2.1
pytest	>=8.3.0
pytest-cov	>=5.0.0
pytest-asyncio	>=0.24.0
pytest-xdist	>=3.6.0
pre-commit	>=3.8.0
ruff	>=0.5.7
ghp-import	>=2.1.0
poetry2conda	>=0.3.0
codecov	^2.1.13
fastapi	^0.115.0

uvicorn	^0.30.0
pydantic	^2.5.0
python-dotenv	^1.0.0
google-cloud-storage	^2.10.0

Table 2: Package Dependencies

Installation

Package installation is recommended to be done through Poetry. First, clone the cephalopy repository from GitHub and set up its development environment. When properly installed, go to your desired console (e.g., PowerShell, Anaconda), navigate to the folder *'cephalopy_package'*, which contains the *pyproject.toml* file. When navigated to that folder, run the following command:

poetry install

This should then read the *pyproject.toml* file, create a dedicated virtual environment, and install all specified dependencies.

Cephalopy Package

The Cephalopy package provides organised code for overall dataset handling, model training, and inference. It is designed for developers and researchers who need to train custom models, integrate systems into existing workflows or extend the underlying algorithms.

For clinical users who only need to analyse cephalograms, it is recommended to use the web application instead, which can be found in the Application section of this User Manual.

The codes are created to ensure easy shareability and to add possibility of local usage through Python. A comprehensive guide is created on the GitHub Documentation², while the data³ and models⁴ specific codes and functionalities have their own extensive (and advanced) documentation.

² GitHub Documentation: https://github.com/BredaUniversityADSAI/2025-26ab-fai3-specialisation-project-team-uwc-landmark-detection/blob/main/cephalopy_package/README.md

³ Package Data Codes Documentation: https://github.com/BredaUniversityADSAI/2025-26ab-fai3-specialisation-project-team-uwc-landmark-detection/blob/main/cephalopy_package/src/cephalopy/data/README.md

⁴ Package Models Functionalities Documentation: https://github.com/BredaUniversityADSAI/2025-26ab-fai3-specialisation-project-team-uwc-landmark-detection/blob/main/cephalopy_package/src/cephalopy/models/README.md

API Endpoints

The API Endpoints can be used through the CLI and API. These endpoints allow for proper and easy usage of functions related to the following topics:

1. Calling, collecting, selecting, and retrieving data, datasets, or models.
2. Performing health checks for data and models.
3. Triggering Model selection, retraining, testing.
4. Performing Predictions (Inference)
5. Generating Analysis reports.

Many of these functionalities are used for the deployed frontend application, making this API critical for the system. The API is currently deployed through Google Cloud Run and can also be used separately from the frontend application to perform these tasks. The API that is currently deployed can be found and used on the following location: <https://ceph-backend-1022645602437.europe-west1.run.app/docs>. If interested what each function does or for usage of the API outside of the application, the documentation for each endpoint is explained below. Deeper explanation on the codes and functions can be found on each respective documentation as specified in the Package Documentation.

Health Checks

Reports on whether the model is loaded, which device is used, the active model's name, and the image size expected by the model.

GET /health

Data Endpoints

Annotation saving

Through this endpoint, it is possible to save annotations to Google Cloud Storage, running the following will save the following 3 folders:

1. X-ray image: data/inference_images/
2. Model's original prediction: data/model_predictions/
3. Final annotation (corrected or approved): data/corrected_annotations/

POST /data/save-annotation

To this, the following Parameters should be added:

- file (UploadFile)
- model_prediction (str)
- final_annotation (str)

- was_corrected (bool)
- ceph_id (str)

Get Data Count

Through this endpoint, you can retrieve the current count of collected images in Google Cloud Storage. This will retrieve a dictionary with the current data count and status.

GET /data/count

Model Endpoints

List Models

List all available model versions from Google Cloud Storage.

GET /models/list

Model Activation

Sets a model as an active production model.

POST /models/{model_id}/activate

Get Active Model

Receives the currently active model.

GET /models/active

Get Model Details

Retrieves detailed information about the specified model.

GET /models/{model_id}

Check New Models

Check if there are any newly trained models that haven't been reviewed yet. This receives all production models, the active model, checks if the new models are retrained or not, and receives respective metadata.

GET /models/new

Clinical Model Comparison

Compare two models with clinical metrics (MRE and SDR@2mm) and proper explanations.

POST /models/compare/clinical

This call expects the following Parameters:

1. request (CompareModelsRequest)
 - a. model_ids (list[str]): two model IDs to compare.

Retraining Endpoints

Data Validation

Check if enough data is available for retraining; there should be at least 50 images available.

POST /retrain/validate

Trigger Retraining

This endpoint triggers a new retraining job, where it first validates data availability, then creates, runs, and returns job.

POST /retrain/trigger

This call expects the following parameters:

- force (*bool*): Skip data validation check
- epochs (*int*)
- learning_rate (*int*)
- batch_size (*int*)

Get Job Status

This endpoint gets the status for the selected retraining job.

GET /retrain/status/{job_id}

Get Retraining History

This endpoint receives the history of recent retraining jobs.

GET /retrain/history

Prediction Endpoints

Predictions from Image

Predicts landmarks from an uploaded image file. This does not automatically save the prediction to Google Cloud Storage, for this the user must store data separately.

POST /predict

This call expects the following Parameters:

- file (*UploadFile*)
- ceph_id (*str*)

Report Generation Endpoints

Generate Document

This endpoint generates a PDF document showing analysis results, notes, and metrics. Combines the previously defined functionality from this file to create the Report Document.

POST /generate_doc

This expects the following Parameters to generate the proper document:

GenerateDocRequest:

- include_title (*bool*)
- main_title (*str*)
- author (*str*)
- patient (*str*)
- descr (*str*)
- logo (*str*)
- title_bg (*str*)
- title_img (*str*)
- titles_li (*list*)
- img_li (*list*)
- img_captions_li (*list[list[str]]*)
- notes_li (*list*)
- metrics_li (*list*)
- path (*str*)
- img_layout (*list[str]*, optional):
 - o "single" (default)
 - o "grid2"
 - o "grid3"

Application

Set-up

The application is deployed through Google Cloud Run. The deployment works through two parts, the deployment of the backend and frontend. The deployment works through Docker. Two Dockerfiles have been made, one for the frontend and one for the backend. A Dockerfile is basically a set of instruction that needs to be followed. These Dockerfiles create a Docker Image, two images are used for the application. These images are created by the following commands in the command prompt:

```
"docker build -t gcr.io/landmark-detection-cepha/ceph-backend ."
```

```
"docker build -t gcr.io/landmark-detection-cepha/ceph-frontend ./frontend"
```

These commands are run in the folder "cephalopy_package" in the main Github folder. After the images are created, they are pushed to Google Cloud Run by the following commands:

```
"gcloud run deploy ceph-backend --image gcr.io/landmark-detection-cepha/ceph-backend --memory 2Gi --cpu 2 --timeout 300 --port 8080 --platform managed --region europe-west1"
```

```
"gcloud run deploy ceph-frontend --image gcr.io/landmark-detection-cepha/ceph-frontend:latest --platform managed --region europe-west1 --allow-unauthenticated --port 8080 --memory 512Mi --cpu 1 --timeout 60"
```

After these commands, both the frontend and backend have been pushed and can be accessed by anyone with the correct link. If updates need to be made to the application, these set of commands can be repeated to deploy the new updates of the application.

Access

The application can be accessed via the following link: <https://ceph-frontend-1022645602437.europe-west1.run.app/>

Functionalities

The application is organized into several tabs, each corresponding to a specific workflow stage.

Landmark Detection

In the Landmark Detection tab, users can upload a cephalometric X-ray image using the upload button, after which the filename is displayed in the interface. By clicking the *Detect Landmarks* button, the uploaded image is passed through the trained deep learning model, which returns predicted landmark positions that are visualised directly on the image.

To support accurate assessment of these predictions, users are provided with several interaction controls. The size of the displayed landmarks can be adjusted using a slider, and the image can be zoomed either through on-screen buttons or mouse-based controls to allow for pixel-level inspection. A contrast slider, ranging from 0% to 200% and centred at 100%, allows users to enhance image visibility. Additionally, users can toggle the visibility of landmark labels as well as custom landmarks, enabling a clearer view of the image when needed.

Landmark Correction

The Landmark Correction tab extends the detection workflow by allowing users to refine and manage landmark predictions. Most quality-of-life features from the Landmark Detection tab are also available here, including zoom, contrast adjustment, landmark resizing, and label toggling.

Within this tab, users can manually correct predicted landmark positions by dragging them directly on the image or using movement controls. If needed, all landmarks can be reset to their original predicted positions using a reset option. The application also supports the creation of custom landmarks, which can be managed through a dedicated interface that allows users to rename or delete these landmarks.

Several saving and export options are available in this tab. Users can export landmark annotations as JSON files, including the original model predictions, manually adjusted landmarks, corrected landmarks, or only custom landmarks. In addition, there is an option to export an image overlay, which saves the original X-ray with the landmarks visually overlaid.

Manual Detection

The Manual Detection tab enables full manual landmark placement without relying on model predictions. Users select landmarks from a predefined list of 19 standard cephalometric landmarks displayed on the left side of the interface and place them on the image by clicking on the desired location. Landmarks can subsequently be repositioned by

dragging them directly on the image or by using a move button in the landmark list. Landmarks may also be deleted in the list if needed.

A field is provided for entering the pixel-to-millimetre conversion rate, which is required for distance-based comparisons and measurements. Once all 19 landmarks have been placed, users can save the annotations as a JSON file formatted specifically for use in model training or retraining pipelines.

The lower section of this tab contains a landmark comparison tool that allows users to compare manually placed landmarks with corrected model predictions. Users can define a distance threshold (with a default of 2 mm), representing a clinically acceptable difference. Landmark differences below this threshold are displayed in green, while those exceeding the threshold are displayed in red. The comparison can be visualised either as a side-by-side view or as a superimposed image. A detailed table below the image shows the manual and corrected coordinates for each landmark, along with the differences in both pixels and millimetres, using colour coding based on the selected threshold.

Cephalometric Analysis

The Cephalometric Analysis tab allows users to perform automated and custom cephalometric analyses using either predicted or manually placed landmarks. Most interaction and visualisation features from previous tabs are also available here.

Users can select the type of analysis to perform, with Steiner analysis set as the default. Custom analyses are also supported, as well as plane-based visualisations that display commonly used cephalometric planes such as the Frankfort Horizontal plane. For the selected analysis, a table is shown containing each measurement, including its value (distance in millimetres or angle in degrees), the corresponding norm, and the deviation from that norm.

When a measurement is selected in the table, it is highlighted and simultaneously visualised on the image, allowing users to clearly see how the measurement is derived. The custom measurement functionality enables users to define new measurements by selecting whether the measurement is linear or angular, choosing the required landmarks (two for linear measurements and three for angular measurements), and entering landmark names either manually or via dropdown menus.

Report Generation

Throughout the process of analysing predictions, it is possible to add notes on any page using the 'Add Notes' button. This button will show up one or more times per page, and the filled in content will correspond with the analysis done on that page. After a note is added, a field will be displayed below the button to display this note.

In the page 'Manual Detection', it is a possibility to add two notes in the page: one for the manual labelling process and one for the comparison to the predicted positions. This page also contains a table that visualises both the manual and predicted coordinates, along with their distance to each other in pixels and millimetres. This table will also be added to the notes.

On the 'Cephalometric Analysis' page, it is possible to select a measurement and add this to the notes. When a measurement is selected, it's row will be shown in a light-blue colour, and the image will visualise the related measurements. After pressing the 'Add Selected Measurement to Notes' button, a list will appear below containing all measurement visualisations that will be added to the report.

Once all notes are added as wanted, you can move to the tab 'Generate PDF', where you will get some further options for the generation of a Report. First, the button 'Show All Notes' will show all notes previously added per page and section to be viewed and edited in one simple overview. Afterwards, there is the option to include a title page or not, along with the possibility to add a title, author, patient identifier, description, and a background to this title page. There is also the option to include the uploaded X-ray on this title page, to help with easier visibility on what cephalogram the report is about. Lastly, there is an image layout option, where you can choose to display images either 1 by 1, 2 by 2 or 3 by 3: this option will display images next to each other in the selected number, helping with either comparing or viewing the images clearly. Finally, when pressing the 'Generate PDF' button, you can generate the PDF Report, of which a preview will be shown below with further options to download this report.

Model Selection

When visiting the 'Models' page on the frontend, an overview of each available model is shown along with their activity and overall performance metrics, being Mean Radial Error (MRE) and Success Detection Rate below 2mm (SDR@2mm). This page displays a timeline graph: Model Performances Over Time, indicating the increase or decrease in overall model performance sorted by date. This graph shows a proper overview of the order of models from old to new along with an easy comparison of the metrics MRE and SDR across

all models. In this page, it is possible to activate the preferred model to be used throughout the Analysis process, this is done using the button: 'Activate Model'.

In the situation where a more detailed comparison between two models is desired, the option 'Compare Models' can be selected from the Models page. Here, you can select the two models to be compared and run a thorough comparison analysis of these models. After detailed documentation of each metric, an overall assessment and clinical interpretation is displayed for each model, showing the MRE accuracy assessment along with a Reliability Assessment by showing how many landmarks are within 2mm threshold. This is then followed with graphs describing SDR comparisons between the models and an overall Performance Profile for specified metrics. Finally, a Detailed Metric Comparison is visualised in a table, containing each metric, their clinical meaning, and a checkmark at the better performing model on specified metric.

Model Retraining

Throughout Cephalometric Analysis, analysed cephalograms can be saved to Google Cloud Storage using the 'Save Prediction' button. This button will then save the image with their correlated predictions, and the later corrections can also be saved through the 'Save to GCS' button on the 'Landmark Detection' page. Once at least 50 cephalogram and annotation combinations are saved, retraining will be made possible. Currently, manual retraining can be performed through the 'Models' page, where clicking the 'Train New Model' button will bring you to the model training page. If this page does not display any data yet, it means that less than 50 images are available for retraining.

The decision to for now only allow manual retraining was made to ensure data quality checks. In this solution, it is important to train the models on high-quality data that can be re-evaluated by medical experts before the models are updated. This will decrease the chance of improper model training or decrease of performance.

Future Recommendations

The current system provides a complete and functional pipeline for automated cephalometric landmark detection and analysis. However, several areas have been identified where further development could improve performance, usability, transparency, and long-term scalability.

Data Recommendations

Future iterations of the system would benefit from the inclusion of additional training data. Expanding the dataset, particularly with more South African patient data, may improve

model generalisation and robustness. The application already supports manual landmark placement and correction, and these user-corrected annotations could be collected and reused as additional labelled data for retraining. In addition, implementing basic data quality checks, such as validating image resolution, image orientation, and annotation completeness, would help ensure consistency in future training data.

Deployment Recommendations

It is recommended to migrate the application and model pipeline to a fully Azure-based cloud deployment. While this approach requires a more complex initial setup and may incur higher costs compared to the current deployment, it enables increased automation and scalability. Azure services could be used to automate model retraining, evaluate and select the best-performing model, and deploy new model versions in a consistent manner. Data quality checks could also be integrated into the retraining pipeline, for example through workflow orchestration tools such as Airflow.

A cloud-based deployment would further support a human-in-the-loop workflow, where user corrections and manual annotations are logged and periodically incorporated into retraining. Over time, this process could allow the model to improve continuously, even when the availability of new raw data is limited.

Application Recommendations

Future application improvements can be divided into two main categories: usability and transparency.

From a usability perspective, extending support for full manual cephalometric tracing would better align the application with existing clinical workflows. Adding batch processing functionality would allow users to analyse multiple cephalograms within a single session. Additional improvements such as session persistence, autosaving, and basic case management could further support longer annotation or analysis workflows.

From a transparency perspective, providing clearer insight into model predictions is recommended. Introducing landmark-level confidence scores would allow users to identify predictions that may require closer inspection. Visualising uncertainty through heatmaps could further support this by indicating areas where the model is less certain. In addition, extending the current comparison functionality to include inter-observer comparisons, such as comparisons between multiple manual annotations, would increase the application's usefulness for research and educational purposes.

These application-level improvements would support more informed user interaction with the system and broaden its applicability beyond automated analysis alone.

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